5.3.28 USB_FS characteristics

Table 69. USB_FS characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDUSB}	USB transceiver operating supply voltage	-	3.0[1]	-	3.6	V
R_{PUI}	Embedded USB_DP pullup value during idle	-	900	-	1575	Ω
R_{PUR}	Embedded USB_DP pullup value during reception	-	1425	-	3090	
Z_{DRV}	Output driver impedance[2]	High and low driver	28	36	44	

1. *USB functionality is ensured down to 2.7 V, but some USB electrical characteristics are degraded in 2.7 to 3.0 V range.*
2. *No external termination series resistors are required on USB_DP (D+) and USB_DM (D-). The matching impedance is already included in the embedded driver.*

5.3.29 Ethernet interface characteristics

Unless otherwise specified, the parameters given in the following Tables are derived from tests performed under the ambient temperature, frcc_c_ck frequency, and V_{DD} supply voltage conditions summarized in Table 16. General operating conditions , with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load Cl = 20 pF
- Measurements points are done at CMOS levels : 0.5 V_{DD}
- IO compensation cell activated

Table 70. Dynamic characteristics: Ethernet MAC signals for SMI

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
t_{MDC}	MDC cycle time(2.5 MHz)	399	400	401	ns
$T_{d(MDIO)}$	Write data valid time	0.5	1	1.5	
$t_{su(MDIO)}$	Read data setup time	14.5	-	-	
$t_{h(MDIO)}$	Read data hold time	0	-	-	

Table 71. Dynamic characteristics: Ethernet MAC signals for RMII

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
$t_{su(RXD)}$	Receive data setup time	2	-	-	ns
$t_{ih(RXD)}$	Receive data hold time	1	-	-	
$t_{su(CRS)}$	Carrier sense setup time	1.5	-	-	
$t_{ih(CRS)}$	Carrier sense hold time	1	-	-	
$t_{d(TXEN)}$	Transmit enable valid delay time	7.5	9.5	14	
$t_{d(TXD)}$	Transmit data valid delay time	7.5	10	15.5	

Table 72. Dynamic characteristics: Ethernet MAC signals for MII

Symbol	Parameter	Min	Typ	Max	Unit
$t_{su(RXD)}$	Receive data setup time	1.5	-	-	ns
$t_{ih(RXD)}$	Receive data hold time	1.5	-	-	
$t_{su(DV)}$	Data valid setup time	1.5	-	-	